Data Analysis for ‘BCMS Prefix Priming’ (No Participant Exclusion)

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About this study

This report analyzes the status of Bosnian/Croatian/Montenegrin/Serbian (BCMS) prefixex in mental grammar using a priming experiment. The work is done by Maša Beslin, Predrag Kovačević, Stefan Ivanović and Utku Turk. The key question is whether lexical prefixes can be individuated as morphemes. We test this by examining whether **prime condition** affects target lexical-decision behavior, and whether that effect remains once we control for trial progression, prime duration, target duration, prime RT, and Inter-Stimulus Interval (ISI).

138 students at the University of Novi Sad, Serbia completed an unmasked auditory continous lexical decision task. Sample size was determined via a priori power analysis to ensure 80% power following [1].

Critical targets are high frequency simplex verbs in the masculine active participle form. Prime-target relations include:

- morphologically and semantically related (predao [gave over] –> dao [gave]),
- morphologically related but semantically opaque (prodao [sold] –> dao [gave]),
- pseudo-morphologically related (nadao [hoped] –> dao [gave]),
- phonologically related (vladao [ruled] –> dao [gave]),
- unrelated controls (preneo [carried over] –> dao [gave])

Participants are assigned to one of four experimental lists, each containing 30 critical item pairs: 5 per priming condition and 10 controls. Each target appears only once per list but is paired with different prime types across lists, enabling within-item comparisons while avoiding repetition effects.

We include 200 filler pairs, which consist of 50% real words and 50% non-words (by making single-segment modifications to existing verbs while respecting BCMS phonotactics). Due to language-specific constraints, the experiment employs a balanced incomplete block design, which accommodates the fact that not all targets can be paired across all conditions, while each condition is equally represented and can be compared to controls [2].

The descriptive plots below first show accuracy, then prime RTs, and then target RTs. The Bayesian model later in the report is fit only to trials with a **correct response to the target**, after the RT trimming steps described below. We incorporate random effects for participants and items to account for within-subject and within-item variability [3]. Our predictors include prime condition, prime duration, target duration, prime Response Time (RT), trial number, and Inter-Stimulus Interval (ISI). Since we are mainly interested in by-condition differences, we treat our numeral predictors as log-transformed and centered (z-scored). Our baselines then are the ‘average’ prime duration, target duration, prime RT, trial number, and ISI.

We employed weakly informative priors as follows. The intercept prior is centered on 6.8 (approximately $e^{6.8} \approx 900$ ms), reflecting a plausible grand-mean log RT. All fixed-effect slopes receive a Normal $(0, 0.5)$ prior, which is skeptical of large effects but permits the data to overwhelm the prior. Random-effect standard deviations and the residual SD use an Exponential (1) prior, which gently regularizes toward smaller variance components. Correlation matrices among random effects are assigned an LKJ (2) prior, which mildly favors lower correlations and helps regularize the maximal random-effect structure.

Table 1: Prior specification for the Bayesian mixed-effects model.

Parameter	Prior	Rationale
Intercept	Normal $(6.8, 0.5)$	Centers on ~900 ms mean RT in log space
Fixed-effect slopes (β)	Normal $(0, 0.5)$	Weakly informative; skeptical of large effects
Random-effect SDs	Exponential (1)	Regularizes toward smaller variance components
Residual SD (σ)	Exponential (1)	Same regularization for observation-level noise
Random-effect correlations	LKJ (2)	Mildly favors lower correlations to stabilize estimation

Model specification

The response variable is **log-transformed target RT**. Prime condition enters the model as a five-level treatment-coded factor with **Control** as the reference level, yielding four contrasts that each compare one priming condition against Control. The remaining fixed effects are continuous nuisance covariates, all z-scored.

Table 2: Model specification summary. The maximal random-effect structure follows R. H. Baayen, D. J. Davidson, and D. M. Bates [3].

Component	Specification
Response	$\log(\text{target RT})$
Fixed effects	Prime condition (treatment-coded, ref = Control), z log prime duration, z log target duration, z log prime RT, z trial number, z ISI
Contrasts	Morpho-Semantic vs Control, Morpho-Opaque vs Control, Pseudo-Morphological vs Control, Phonological vs Control
By-subject random effects	Intercept + all fixed-effect slopes
By-item random effects	Intercept + all fixed-effect slopes
Family	Gaussian
Chains / Iterations	4 chains × 6 000 iterations (2 000 warmup)
Sampler settings	`adapt_delta` = 0.99, `max_treedepth` = 15
Backend	CmdStanR

Results

Accuracy

Accuracy is shown first because it gives the clearest screening view of whether any prime condition is associated with a speed-accuracy tradeoff. The y-axis is expressed as a percentage rather than a proportion.

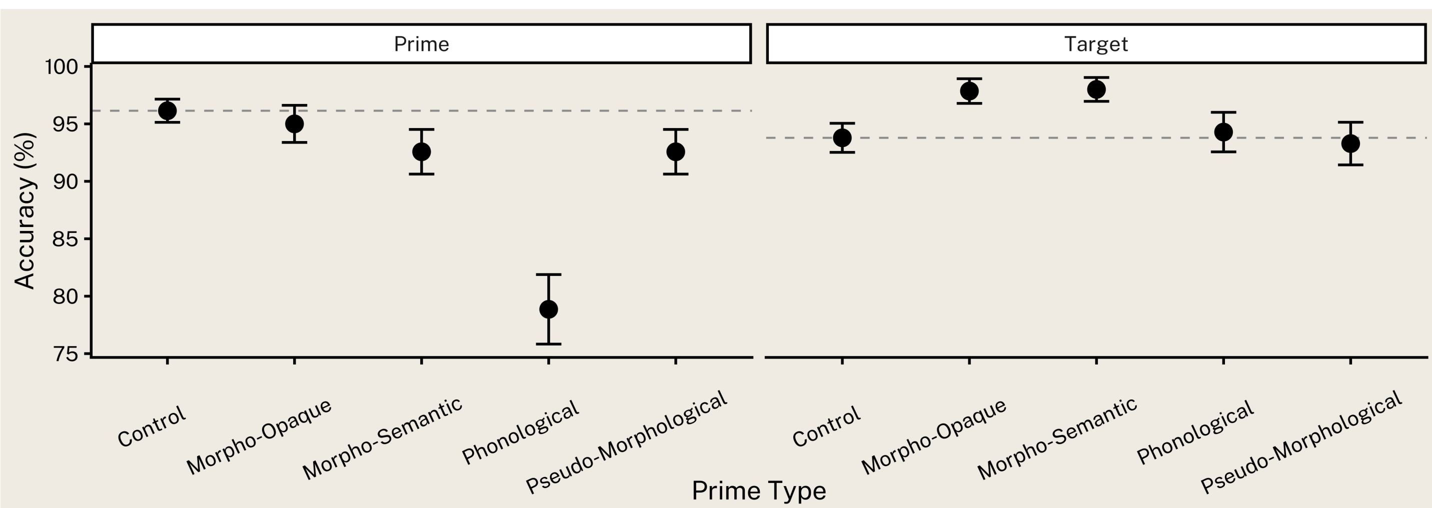

Figure 1: Prime and target accuracy by prime condition. Error bars show 95% confidence intervals. The dashed line marks the Control-condition mean.

prime	pword	acc	n
kuluchio	word	0.0937500	32
navodao	word	0.3437500	32
presitnio	word	0.4864865	37
domogao	word	0.5625000	32
nakapao	word	0.5789474	38

prime	pword	acc	n
naveslao	word	0.5945946	37

target	tword	acc	n
ticao	word	0.4214286	140

In this version of the analysis, no participants were excluded based on accuracy. Trials containing a prime or target with an item-level accuracy below 60% were excluded (8.29 % of the data). Trials with incorrect responses to targets were discarded (2.86 % of the remaining data). Subsequently, trials with outlier RTs (target RT < 250 ms or > 2500 ms; prime RT < 250 ms or > 2000 ms) were removed, excluding an additional 9.01 % of the remaining data. Overall, 18.93 % of the original experimental trials were removed, leaving 3405 trials for the final analysis.

Prime RTs

The next figure summarizes prime RTs after restricting the timing data to target-correct trials and trimming outliers. The dashed horizontal line shows the mean RT in the Control condition, so each condition can be read relative to that baseline.

Prime condition	N	Mean prime RT (ms)	SE
Control	1195	1286.80	7.28
Morpho-Opaque	627	1322.18	10.40
Morpho-Semantic	597	1310.21	9.68
Phonological	458	1338.10	12.61
Pseudo-Morphological	528	1295.33	10.79

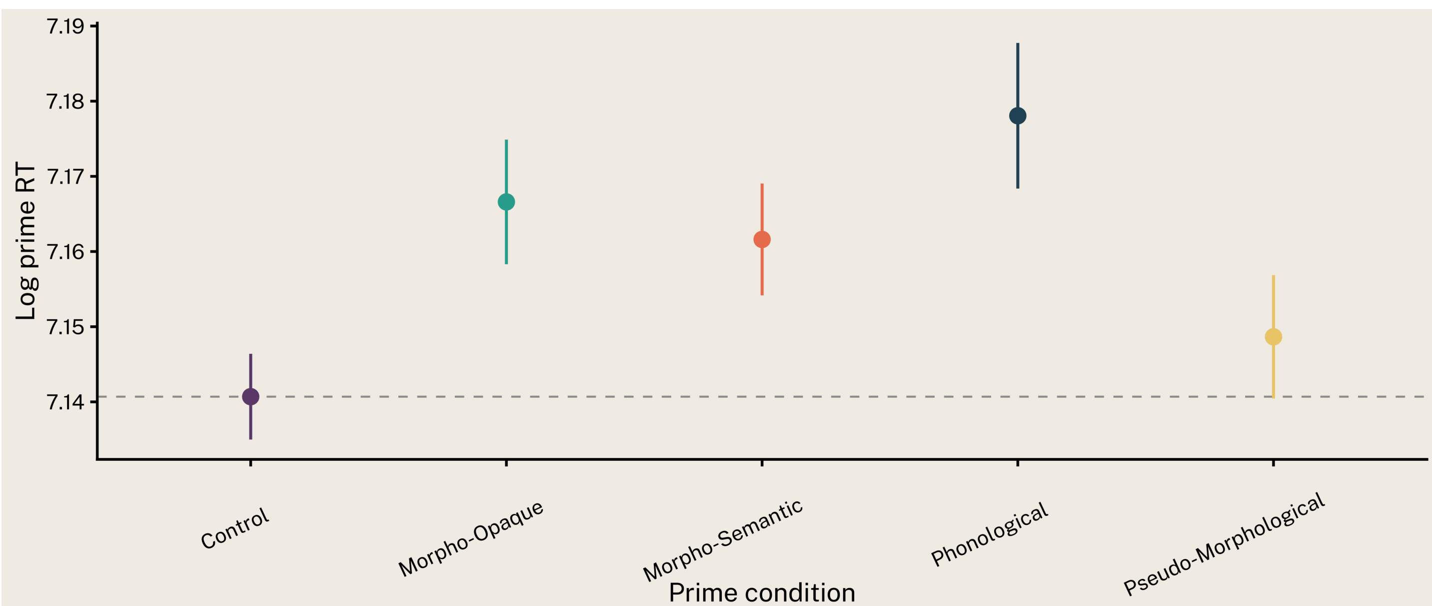

Figure 2: Prime RTs by prime condition, using only target-correct trials after RT trimming. The dashed line marks the Control-condition mean.

Descriptively, mean prime RTs were similar across conditions: Control 1287 ms (SE = 7), Morpho-Semantic 1310 ms (SE = 10), Morpho-Opaque 1322 ms (SE = 10), Pseudo-Morphological 1295 ms (SE = 11), and Phonological 1338 ms (SE = 13). A simple linear mixed-effects model predicting log prime RT from prime condition with random intercepts for subjects and items showed a significant main effect of condition, $F(4, 2348.8) = 7.62$, $p < .001$. This suggests that prime processing difficulty varied across conditions, which motivates the inclusion of prime RT as a covariate in the target RT model. Regardless, we include log prime RT as a z-scored covariate in our main target RT model to account for by-trial variation in processing speed, ensuring that any condition effects on target RTs are not confounded with trial-level speed differences.

Target RTs

Target RTs are plotted on the log scale because the model is fit to log-transformed target RTs. The dashed horizontal line again marks the Control-condition mean, so the figure lines up with the model coefficients that compare each condition to Control.

Prime condition	N	Mean log RT	SE
Control	1195	7.09	0.01
Morpho-Semantic	597	7.03	0.01
Morpho-Opaque	627	7.03	0.01
Pseudo-Morphological	528	7.02	0.01
Phonological	458	7.06	0.01

Mean target RTs by condition were: Control 1237 ms (SE = 8; log: 7.095), Morpho-Semantic 1150 ms (SE = 10; log: 7.027), Morpho-Opaque 1164 ms (SE = 11; log: 7.034), Pseudo-Morphological 1142 ms (SE = 11; log: 7.017), and Phonological 1192 ms (SE = 13; log: 7.059). Numerically, Pseudo-Morphological elicited the fastest responses and Control the slowest, with a range of 95 ms across conditions.

Compared to the Control condition, all morphologically related conditions showed descriptively faster target responses: Morpho-Semantic was 87 ms faster (Δ log = 0.068), Morpho-Opaque was 73 ms faster (Δ log = 0.061), and Pseudo-Morphological was 95 ms faster (Δ log = 0.078). The Phonological condition was 45 ms faster (Δ log = 0.036). These descriptive differences are consistent with a facilitation effect for morphological priming, though formal inference requires the model reported below. After excluding trials with missing duration values, the modeling dataset contains 3405 observations from 138 participants across 30 items.

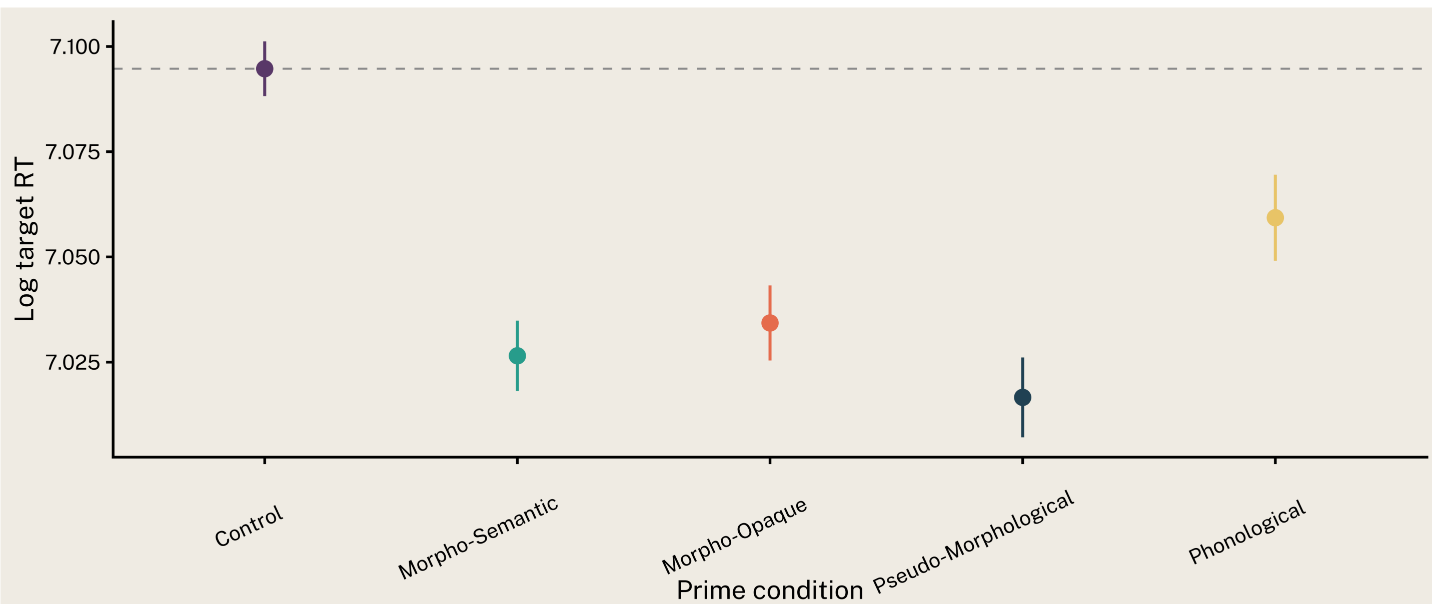

Figure 3: Log-transformed target RTs by prime condition. Boxes show the condition distribution; dots show individual observations with low alpha. The dashed line marks the Control-condition mean.

Statistical analysis

The Bayesian model below uses only the cleaned timing dataset: trials with a correct response to the target, plus RT trimming for both prime and target. The outcome is **log target RT**, and the predictors are prime condition, z-scored trial number, z-scored log prime duration, z-scored target duration, z-scored log prime RT, and z-scored ISI.

parameter	β	SE	CI	P(β > 0)
Morpho-Semantic vs Control	−0.095	0.015	[−0.125, −0.064]	< 0.001
Morpho-Opaque vs Control	−0.074	0.016	[−0.105, −0.042]	< 0.001
Pseudo-Morphological vs Control	−0.069	0.019	[−0.107, −0.033]	< 0.001
Phonological vs Control	−0.053	0.024	[−0.101, −0.006]	0.015

parameter	β	SE	CI	P(β > 0)
Trial number	0.013	0.007	[−0.001, 0.028]	0.970
Prime duration	−0.018	0.008	[−0.035, −0.001]	0.018
Target duration	0.033	0.009	[0.016, 0.050]	> 0.999
Prime RT	0.084	0.006	[0.072, 0.095]	> 0.999
ISI	−0.001	0.003	[−0.008, 0.005]	0.327

The coefficient table summarizes how each predictor shifts log target RT while holding the others constant. For the condition terms, the reference level is Control, so negative coefficients under β column correspond to faster target responses relative to Control and positive coefficients correspond to slower target responses.

The fixed-effect results can be read in terms of posterior direction probabilities under the P(β > 0) column. Values near 0.50 indicate substantial posterior uncertainty, whereas values near 1.00 or 0.00 indicate that the posterior mass is concentrated on one side of zero.

- Phonological vs Control: β = −0.053, P(β < 0) = 0.985, CI [−0.101, −0.006].
- Prime duration: β = −0.018, P(β < 0) = 0.982, CI [−0.035, −0.001].
- Trial number: β = 0.013, P(β > 0) = 0.970, CI [−0.001, 0.028].

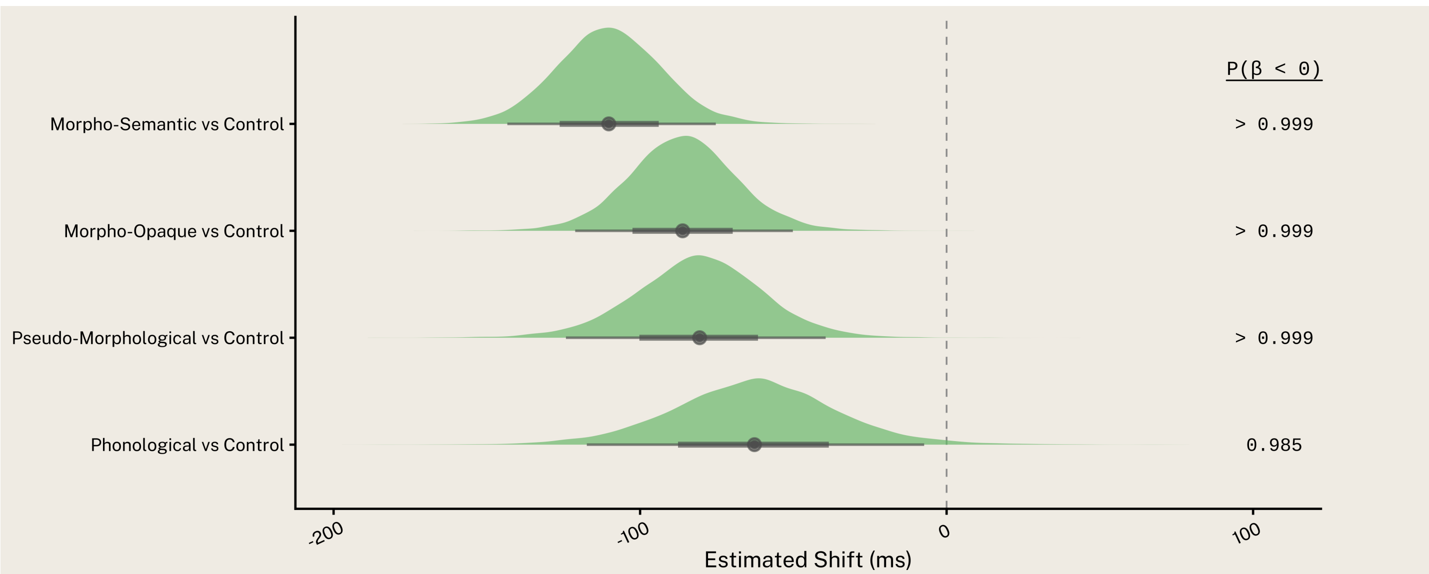

Figure 4: Posterior distributions for condition effects. Values are transformed to estimated millisecond shifts relative to the grand mean intercept. Green slabs indicate facilitation; red slabs indicate inhibition.

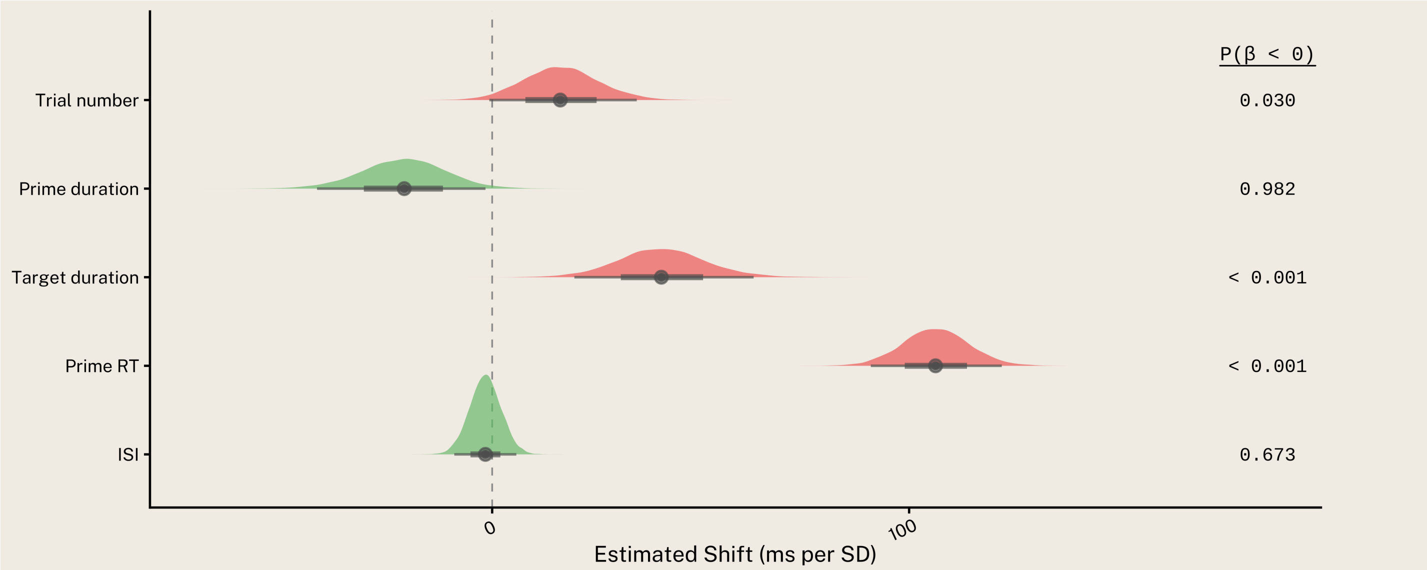

Figure 5: Posterior distributions for controlling variables. Values represent the millisecond shift associated with a 1 SD change in the predictor.

contrast	Difference (ms)	SE	CI	direction_prob
Morpho-Semantic vs. Morpho-Opaque	−23.93	22.05	[−67.32, 19.31]	P(β < 0) = 0.866
Morpho-Semantic vs. Pseudo-Morphological	−29.13	25.49	[−78.28, 21.49]	P(β < 0) = 0.878
Morpho-Semantic vs. Phonological	−47.16	30.09	[−106.94, 11.79]	P(β < 0) = 0.946
Morpho-Opaque vs. Pseudo-Morphological	−5.20	26.50	[−55.63, 48.18]	P(β < 0) = 0.589
Morpho-Opaque vs. Phonological	−23.23	30.61	[−84.25, 36.60]	P(β < 0) = 0.782
Pseudo-Morphological vs. Phonological	−18.03	32.43	[−82.67, 46.51]	P(β < 0) = 0.724

The contrast table compares the non-control conditions directly. Each row tests whether the condition named on the left has a larger coefficient than the condition named on the right, and the direction-probability column states whether the posterior favors a positive or negative difference.

These pairwise contrasts show whether the non-control prime conditions differ from each other, rather than only from Control.

- Morpho-Semantic vs. Morpho-Opaque: estimated difference = , P(β < 0) = 0.866, CI [−67.32, 19.31].
- Morpho-Semantic vs. Pseudo-Morphological: estimated difference = , P(β < 0) = 0.878, CI [−78.28, 21.49].
- Morpho-Semantic vs. Phonological: estimated difference = , P(β < 0) = 0.946, CI [−106.94, 11.79].

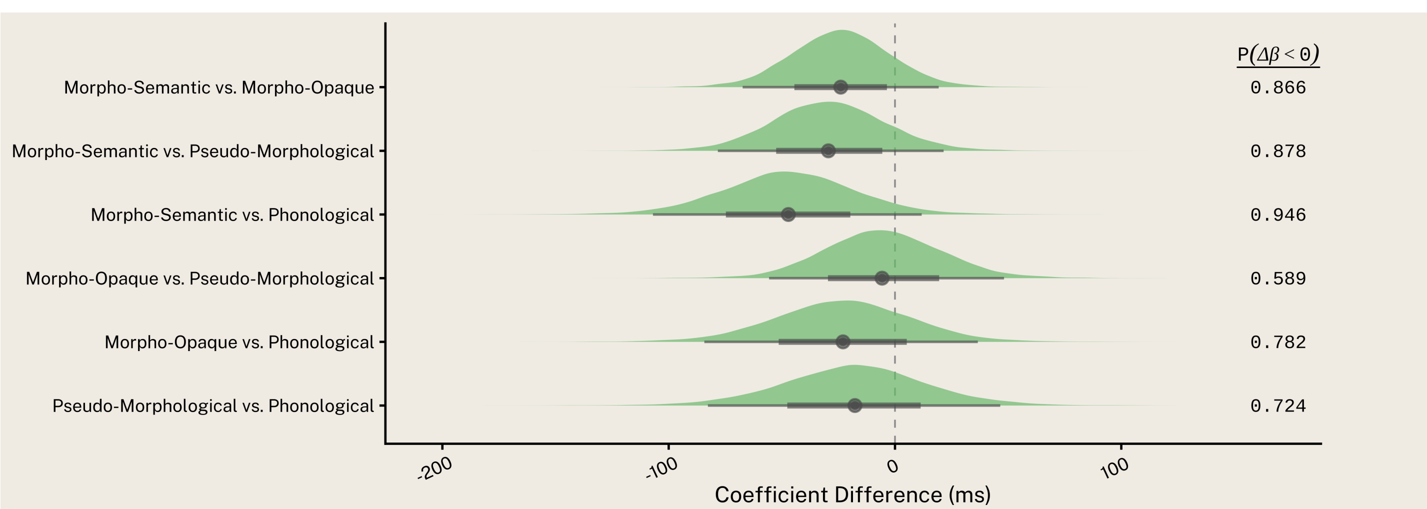

Figure 6: Posterior distributions for pairwise differences among non-control prime conditions, transformed to milliseconds. Green slabs indicate a negative difference (left condition faster); red slabs indicate a positive difference (left condition slower).

contrast	β diff	SE	CI	direction_prob
Genuine prefix vs. Pseudo-Morphological	−0.015	0.021	[−0.055, 0.027]	P(β < 0) = 0.776
Surface segmentable vs. Morpho-Semantic	0.024	0.018	[−0.011, 0.058]	P(β > 0) = 0.913

These composite contrasts ask whether Morpho-Opaque behaves more like the genuinely prefixed condition (Morpho-Semantic) or the surface-segmentable condition (Pseudo-Morphological). If only the first contrast is credible, MO patterns with real morphology; if only the second, MO patterns with pseudo-morphology; if both or neither, MO occupies an intermediate position.

- Genuine prefix vs. Pseudo-Morphological: estimated difference = −0.015, P(β < 0) = 0.776, CI [−0.055, 0.027].
- Surface segmentable vs. Morpho-Semantic: estimated difference = 0.024, P(β > 0) = 0.913, CI [−0.011, 0.058].

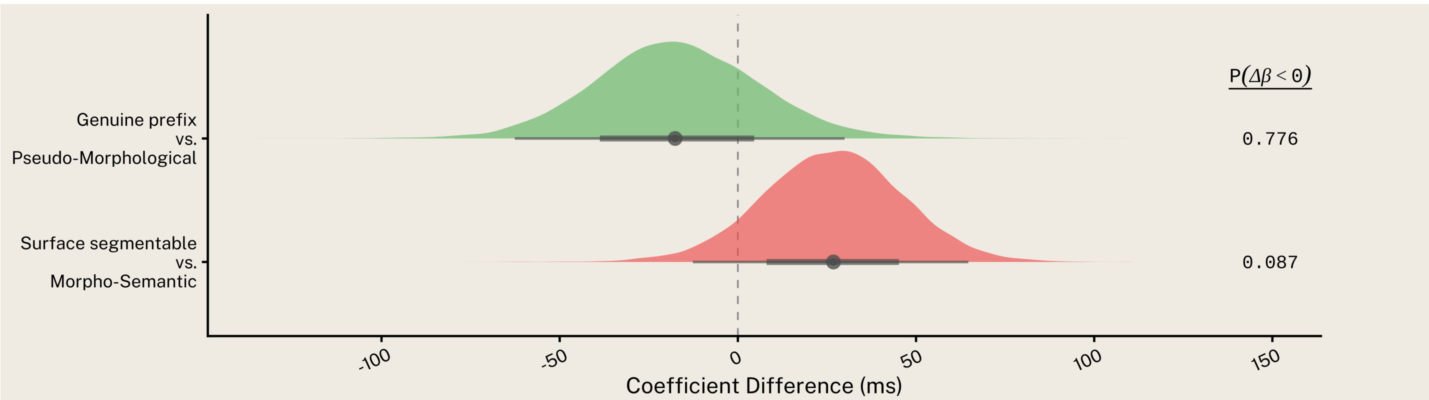

Figure 7: Posterior distributions for composite contrasts in milliseconds. Green slabs indicate a negative difference (left cluster faster); red slabs indicate a positive difference (left cluster slower).

Bibliography

[1] S. Vasishth, H. Yadav, D. J. Schad, and B. Nicenboim, “Sample Size Determination for Bayesian Hierarchical Models Commonly Used in Psycholinguistics,” *Computational Brain & Behavior*, vol. 6, no. 1, pp. 102–126, 2023.

[2] R. E. Bechhofer and A. C. Tamhane, “Incomplete Block Designs for Comparing Treatments with a Control: General Theory,” *Technometrics*, vol. 23, no. 1, pp. 45–57, 1981.

[3] R. H. Baayen, D. J. Davidson, and D. M. Bates, “Mixed-Effects Modeling with Crossed Random Effects for Subjects and Items,” *Journal of Memory and Language*, vol. 59, no. 4, pp. 390–412, 2008.